Probability - 2 (cont.)

From last topic

$$p(t \mid c \wedge x) = ?$$

$$= \frac{p(t \wedge c \wedge x)}{p(c \wedge x)}$$

$$= \frac{0.108}{0.108 + 0.072}$$

$$= \frac{0.108}{0.18} = 0.6$$

Next topic - Independence

- Full Independence
- Conditional Independence

Let's talk about full-independence. (FI)

what happens in conditional probability?

$P(A|B)$.

FI means outcome of A, doesn't affect outcome of B. or, vice-versa.

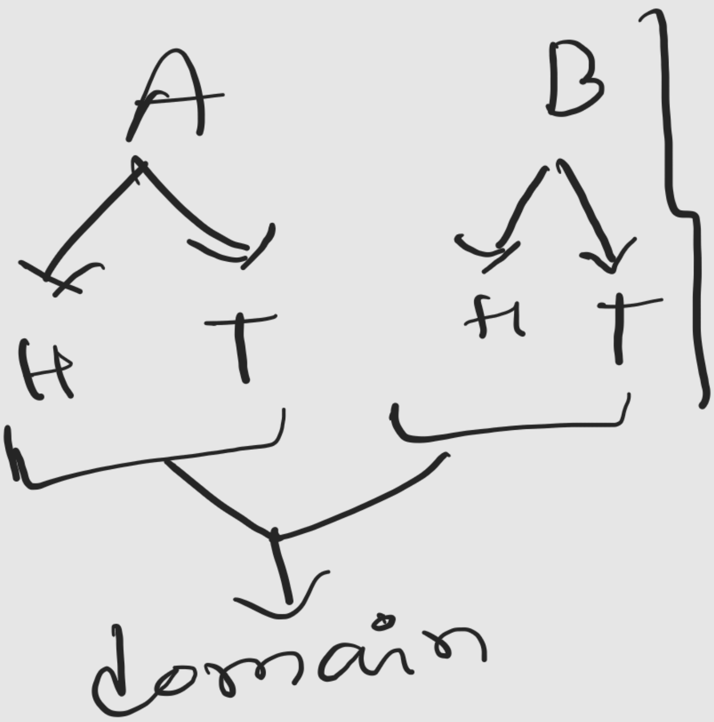
Example: multiple coin toss. (fair coin)

<u>Formula of FI:</u>

$$P(A \wedge B) = P(A) * P(B)$$

Let's say 1st toss – event A

2nd toss – event B



HH $\Rightarrow P = 0.25$

HT

TH all same

TT

$\Rightarrow p(H \wedge T) = ? = 0.25$

now,

$$p(H \wedge T) = p(H) * p(T)$$

$$\Rightarrow 0.25 = 0.5 \times 0.5$$

$$\Rightarrow 0.25 = 0.25$$

So, coin tosses are independent

now, conditional Independence (CI)

Example from online:

Person-A & Person-B has gone to a restaurant.

You know, if there's a thunderstorm, they might get home late.
You don't know initially there's a storm taking place or not. now, if Person-A is late.

you will assume there is a storm happening & person-B will be late as well. Here, A's being late affects personB's being late.

So, they are not independent as person-A's outcome is affecting person-B's outcome.

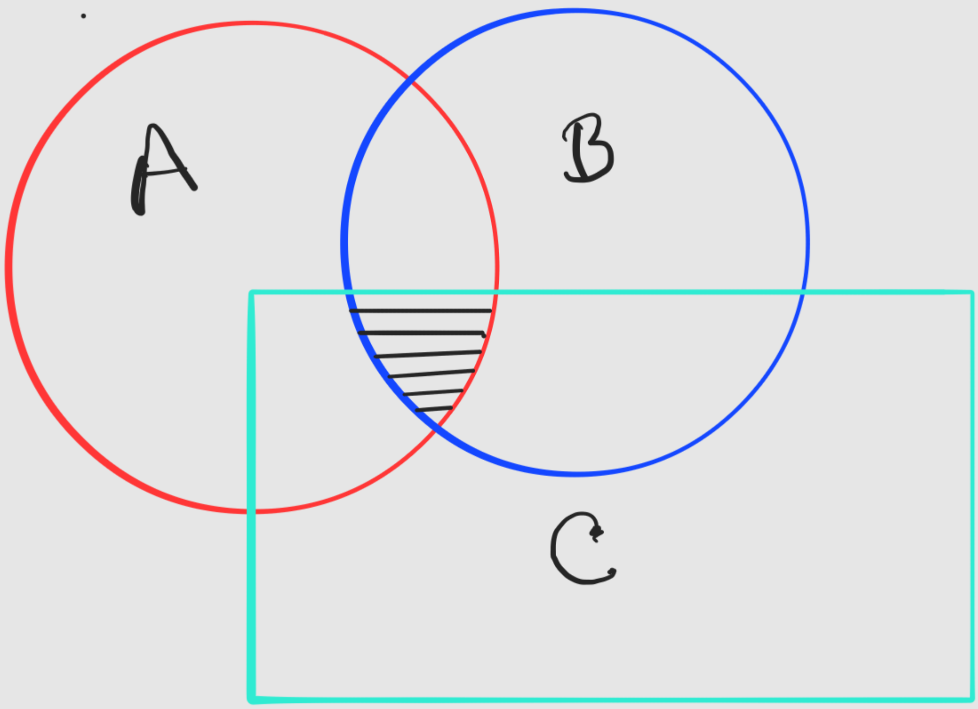
Now, say you know already storm in happening.

Then, you will know both of them (A & B) have a high chance of being late. Here, whether person A is late or not, that will not affect the probability of person B being late.

Now, then person-A & person-B are independent given event C / condition.

Formula of CI:

$$P(A \wedge B \mid C) = P(A \mid C) * P(B \mid C)$$



A, B and C will have an intersection.

However, if A & B are independent,

it doesn't gurantee that they will be conditionally independent.

The formula can be written differently.

$$P(A \wedge B | C) = P(A|C) * P(B|C)$$

$$\Rightarrow \frac{P(A \wedge B \wedge C)}{P(C)} = P(A|C) * P(B|C)$$

$$\Rightarrow P(A \wedge B \wedge C) = P(C) * P(A|C) * P(B|C)$$

We will use the first one for the maths in today's class.

$P(sm \wedge st \wedge pr)$	sm		¬sm	
	st	¬st	st	¬st
pr	0.432	0.16	0.084	0.008
¬pr	0.048	0.16	0.036	0.072

Q1. Is sm independent of st?

ans: sm = A, st = B

$$P(A \wedge B) = P(A) * P(B)$$

$$\Rightarrow 0.432 + 0.048 = (0.432 + 0.048 + 0.16 + 0.16) * (0.432 + 0.048 + 0.084 + 0.036)$$

$\Rightarrow 0.48 = 0.8 \times 0.6$

$\Rightarrow 0.48 = 0.48$

So, sm is independent of st.

Q_2. Is pr independent of st? try yourself

<u>now, CI math</u>

Q_1. Is sm conditionally independent of pr, given st?

$\Rightarrow P(sm \wedge pr \mid st) = P(sm \mid st) * P(pr \mid st)$

$$E.S = P(sm \wedge pr \mid st)$$

$$= \frac{P(sm \wedge pr \wedge st)}{P(st)}$$

$$= \frac{0.432}{0.432 + 0.048 + 0.084 + 0.036}$$

$$= \frac{0.432}{0.36}$$

$$= 0.72$$

$$R.S = P(sm \mid st) * P(pr \mid st)$$

$$= \frac{P(sm \wedge st)}{P(st)} * \frac{P(pr \wedge st)}{P(st)}$$

$$= \frac{P(Sm \wedge St) * P(Pr \wedge St)}{P(St)^2}$$

$$= \frac{(0.432+0.048) \times (0.432+0.084)}{(0.6)^2}$$

$$= \frac{0.48 \times 0.516}{0.36}$$

$$= 0.688$$

So, $L.L \neq R.S$. Hence Sm is not conditionally independent of Pr, given St.

$[0.72 \neq 0.71999]$

Baye's theorem

we knew,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \cdots (1)$$

similary,

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$\Rightarrow P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(B|A)\,P(A) = P(A \cap B) \quad \cdots (2)$$

from ean (1)

$$\boxed{P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}}$$

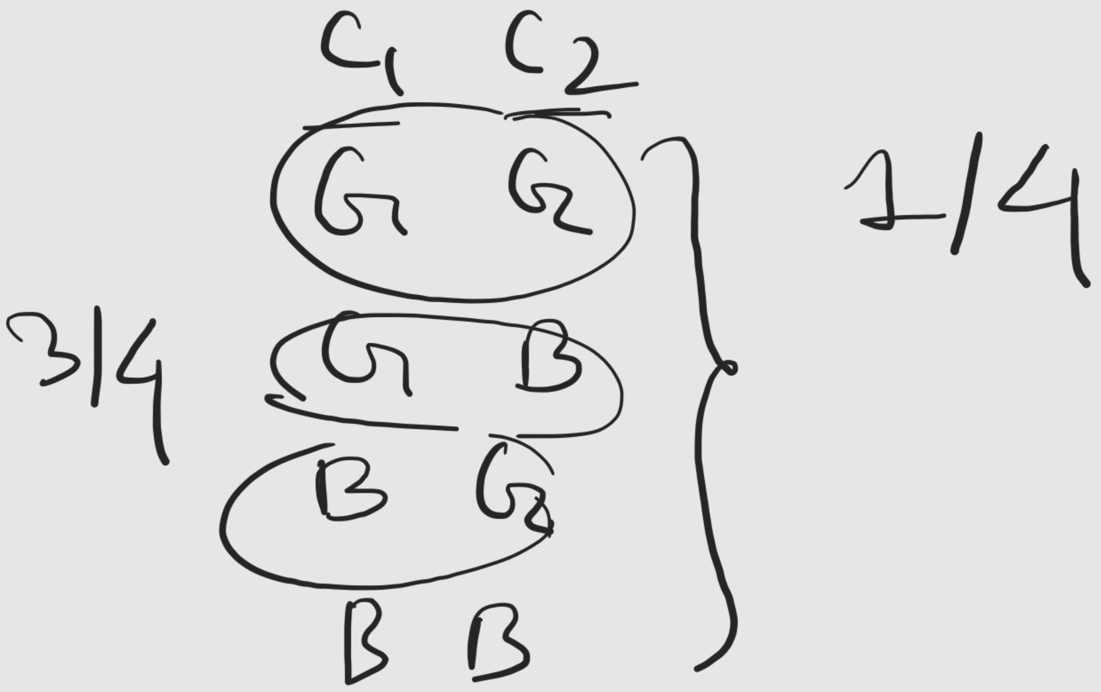
Q1. A couple has two children. Given that one them is a girl, what is the probability that both are going to be a girl?

ans:

$$P(2G \mid 1G) = \frac{P(2G \cap 1G)}{P(1G)}$$

$$= \frac{P(1G \mid 2G) \cdot P(2G)}{P(1G)}$$

$$= \frac{1 \cdot 1/4}{3/4} = 1/3$$



Q_2. We want to compute the probability that it will rain tomorrow, given that the sky is cloudy.

rain – R, cloudy – C

$p(R) = 0.3$

$p(C|R) = 0.8$

$P(C) = 0.6$

$$P(R|C) = ?$$

$$\Rightarrow P(R|C) = \frac{P(R \cap C)}{P(C)}$$

We don't have $P(R \cap C)$.

$$\Rightarrow P(R|C) = \frac{P(C|R) \cdot P(R)}{P(C)}$$

$$= \frac{0.8 * 0.3}{0.6}$$

$$= 0.40$$